Almost everywhere divergence of polynomial interpolation with sublinear excess degree

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Abstract

Let X_n be an arbitrary set of n distinct points in $[-1,1]$, and let r_n be nonnegative integers with $r_n = o(n)$. We prove that there is a real continuous function f such that every sequence of polynomials p_n, of degree at most $n+r_n$, satisfying $p_n = f$ on X_n, has $\limsup_{n\to\infty} |p_n(x)| = \infty$ for almost every $x \in [-1,1]$. The exceptional null set may depend on the sequence (p_n). This extends the Erdős–Vértesi theorem on almost-everywhere unboundedness of Lagrange interpolation for arbitrary node arrays to nonunique interpolation with $o(n)$ excess degree, and in particular answers Erdős Problem 1152 in a stronger form. The proof combines localized functions in Bernstein spaces with uniform asymptotics for weighted Christoffel–Darboux kernels. A finite interpolation construction and the Baire category theorem give a single function valid for every admissible interpolation sequence.

1 Introduction

Write Π_d for the space of real polynomials of degree at most d. All measures of subsets of $[-1,1]$ below are Lebesgue measures. The uniform norm on this interval is denoted by $\|\cdot\|_\infty$.

Theorem 1. *For each $n \geq 1$, let $X_n \subset [-1,1]$ consist of n distinct points. Suppose that $r_n \geq 0$ are integers and $r_n/n \to 0$. There exists $f \in C([-1,1];\mathbb{R})$ such that every sequence satisfying*

$$p_n \in \Pi_{n+r_n}, \qquad p_n(x) = f(x) \quad (x \in X_n)$$

also satisfies

$$\limsup_{n\to\infty} |p_n(x)| = \infty \quad \textit{for almost every } x \in [-1,1]. \tag{1.1}$$

Erdős and Vértesi proved that for every triangular array of distinct nodes in $[-1,1]$, there is a continuous function whose Lagrange interpolation polynomials are unbounded almost everywhere [1, 2]. Theorem 1 extends this arbitrary-node result to nonunique interpolation: degrees up to $n+r_n$, with $r_n = o(n)$, are allowed, and the same continuous function works for every admissible sequence of interpolants. This also answers the question in [4, Problem 2.42], recorded as Erdős Problem 1152 [5], in the stronger form (1.1). Indeed, for the degree condition $\deg p_n < (1+\varepsilon_n)n$, set $r_n = \max\{0, \lceil \varepsilon_n n \rceil - 1\}$; then $r_n = o(n)$ and $\deg p_n \leq n + r_n$.

The contrast with fixed positive relative excess is provided by Erdős, Kroó, and Szabados [3]: for suitable node arrays and each fixed $\varepsilon > 0$, every continuous function admits uniformly convergent interpolants of degree at most $\lfloor (1+\varepsilon)n \rfloor$. For background on divergence phenomena in Lagrange interpolation, see Szabados and Vértesi [6].

The two analytic ingredients are the construction in Olevskii and Ulanovskii [9, Lemma 4.1] and the uniform kernel asymptotics of Kriecherbauer, Schubert, Schüler, and Venker [8, Theorem 1.3 and Theorem 1.8 (bulk universality)]. Section 3 records the kernel estimates in the normalization used below, while Section 6 derives the corresponding local external field from a weak limit of the node measures. We first obtain interpolation data that force large values on a fixed fraction of a local interval, uniformly over all admissible polynomial corrections. We then combine finitely many such data sets to rule out common sets of bounded values. The final step is a category argument in $C([-1,1];\mathbb{R})$.

2 Localized Bernstein functions and polynomial corrections

For $\sigma > 0$, let B_σ denote the space of entire functions of exponential type at most σ that are bounded on $\mathbb{R}$. Norms of functions in B_σ are taken on $\mathbb{R}$.

Lemma 2. *There are absolute constants $c, C > 0$ with the following property. For every sufficiently large integer ρ, put*

$$R_0 = \rho - \sqrt{\rho}, \qquad \gamma = R_0^{-1/9}.$$

For every finite set $\Gamma \subset [-\rho, \rho)$ with $\#\Gamma \le 2\rho + 1$, there is an entire function g, real on $\mathbb{R}$, such that

$$g \in B_{\pi-4\gamma}, \qquad \max_{\Gamma} |g| \le 1, \qquad \|g\|_\infty \le C \log \rho, \tag{2.1}$$

$$g(u_0) \ge c \log \rho \quad \text{for some } |u_0| \le R_0, \tag{2.2}$$

$$|g(u)| \le C(|u| - R_0)^{-3} \quad (|u| \ge \rho). \tag{2.3}$$

For each fixed ρ, these choices $g = g_\Gamma$ satisfy

$$\sup_{\Gamma} \|g_\Gamma\|_{L^1(\mathbb{R})} \le C_\rho, \qquad \sup_{\Gamma} \int_{|u|>T} |g_\Gamma(u)|\, du \le \frac{C}{(T - R_0)^2} \quad (T \ge \rho),$$

where the suprema range over the finite sets Γ specified above. There is also an interval, of length bounded below by an absolute positive constant, contained in $(-\rho, \rho) \setminus \Gamma$, on which $g \ge c' \log \rho$, with an absolute constant $c' > 0$.

Proof. Adapt the construction of [9, Lemma 4.1, pp. 184–185] by reducing its initial bandwidth from $\pi - 4R_0^{-1/9}$ to $\pi - 8R_0^{-1/9}$. The fourth-power sinc factor below then leaves the required margin 4γ below π. Put $w = \pi - 8\gamma$ and

$$\Lambda = \Gamma + 2R_0\mathbb{Z}.$$

Since Γ is finite, Λ is uniformly discrete. Being periodic, its lower and upper uniform densities coincide; write their common value as D. Then

$$D \le \frac{2\rho + 1}{2R_0} \le 1 + CR_0^{-1/2}.$$

Write $K_s(\Lambda, B_w)$ for the sampling constant. If Λ is a sampling set for B_w, then [9, Theorem C(i), p. 179] gives $D > w/\pi$. Hence $D\Lambda$ has density one and [9, Theorem 1(i), p. 180] gives

$$K_s(\Lambda, B_w) = K_s(D\Lambda, B_{w/D}) \ge C \log \frac{\pi}{\pi - w/D} \ge c \log R_0,$$

since, for large ρ,

$$0 < \pi - \frac{w}{D} = \frac{\pi D - w}{D} \le C\big(R_0^{-1/2} + \gamma\big) \le C\gamma.$$

If Λ is not a sampling set, its sampling constant is infinite, so the same lower bound is immediate. Thus one can choose $g_1 \in B_w$, bounded by one on Λ, whose norm is a fixed constant times $\log R_0$. Choose a point at which its absolute value is at least half this norm. A translation by a multiple of $2R_0$ places this point at $u_0 \in [-R_0, R_0]$, without changing the bound on Λ. Multiply by a constant of modulus one so that $g_1(u_0) > 0$, then replace $g_1(z)$ by $(g_1(z) + \overline{g_1(\bar{z})})/2$. This entire function is real on $\mathbb{R}$, retains the value at u_0, and satisfies the same bandwidth and upper bounds.

Multiply this function by

$$\left(\frac{\sin(\gamma(u-u_0))}{\gamma(u-u_0)}\right)^4.$$

The resulting function has bandwidth at most $\pi - 4\gamma$ and retains the node bound and the value at u_0. For $|u| \ge \rho$, the bound

$$C \log R_0\, \gamma^{-4}(|u| - R_0)^{-4} \le C(|u| - R_0)^{-3}$$

follows from $\log R_0\, R_0^{4/9}/\sqrt{\rho} = O(1)$. This proves (2.1)–(2.3) and the assertions about integrability. Finally, Bernstein's inequality gives $\|g'\|_\infty \le \pi C \log \rho$. Thus the peak contains an interval of fixed positive length on which $g \ge c' \log \rho > 1$. This interval contains no point of Γ and lies inside the original box. $\square$

For a finite set $Y \subset \mathbb{R}$, define

$$P_Y(x) = \prod_{y \in Y} (x - y).$$

Lemma 3 (Alternating peaks)**.** *Let Y consist of $N \ge 1$ distinct real points, let $H > 0$, and let $d \ge 0$ be an integer. Suppose that disjoint intervals $J_1, \ldots, J_K$, in increasing order and missing Y, and polynomials $B_1, \ldots, B_K \in \Pi_{N-1}$ satisfy*

$$\max_{y \in Y} \sum_j |B_j(y)| \le 2, \qquad B_j(x) > 2H + \sum_{i \ne j} |B_i(x)| \quad (x \in J_j).$$

There are data $v_y \in [-1, 1]$ on Y such that every $p \in \Pi_{N+d}$ interpolating these data satisfies $|p| > H$ throughout at least $(K - d - 1)/2$ of the intervals J_j.

Proof. Choose signs ε_j so that $\operatorname{sgn}(\varepsilon_j/P_Y)|_{J_j} = (-1)^j$, and put $b = \frac{1}{2}\sum_j \varepsilon_j B_j$. Then $|b| \le 1$ on Y, while peak dominance gives

$$|b| > H, \qquad \operatorname{sgn}(b/P_Y)|_{J_j} = (-1)^j. \tag{2.4}$$

Set $v_y = b(y)$. Any interpolant in Π_{N+d} has the form $p = b + P_Y q$ with $\deg q \le d$. Call J_j a low interval if it contains a point x_j with $|p(x_j)| \le H$. At such a point the sign of $q(x_j)$ is opposite to that of $b(x_j)/P_Y(x_j)$. If there are B low intervals, the resulting signs alternate at least $2B - K - 1$ times. Indeed, deleting $K - B$ terms from a fully alternating sequence removes at most $2(K - B)$ sign changes. Consequently $2B - K - 1 \le d$, so

$$B \le \frac{K + d + 1}{2}. \tag{2.5}$$

The asserted bound follows. $\square$

3 An explicit external field and its polynomial kernel

For a finite positive measure μ, use the logarithmic-potential convention

$$V_\mu(x) = \int \log|x-y|\, d\mu(y).$$

Define

$$Q(u) = \frac{1}{2}\big((1+u)\log(1+u) + (1-u)\log(1-u)\big), \qquad |u| < 1. \tag{3.1}$$

Then $Q(0) = 0$, $Q'(u) = \operatorname{arctanh} u$, and $Q''(u) = (1-u^2)^{-1}$.

Lemma 4. *For $0 < t < 1$, set $a_t = \sqrt{1-(1-t)^2}$. The equilibrium measure of mass t in the external field Q on $[-1,1]$ has density*

$$\rho_t(u) = \frac{1}{\pi}\arccos\frac{1-t}{\sqrt{1-u^2}}\,\mathbf{1}_{\{|u|<a_t\}}. \tag{3.2}$$

Thus $Q - V_{\rho_t}$ is constant on $[-a_t, a_t]$ and is no smaller elsewhere in $[-1,1]$.

Proof. Differentiation with respect to t, away from $|u| = a_t$, gives

$$\partial_t \rho_t(u) = \frac{\mathbf{1}_{\{|u|<a_t\}}}{\pi\sqrt{a_t^2 - u^2}}. \tag{3.3}$$

This is the arcsine probability density on $[-a_t, a_t]$, so ρ_t has total mass t. Its Cauchy transform can be computed by integrating the transforms of these arcsine measures. For $0 < x < a_t$, its real boundary value is

$$\int_0^{1-\sqrt{1-x^2}} \frac{ds}{\sqrt{x^2 - a_s^2}} = \operatorname{arctanh} x = Q'(x). \tag{3.4}$$

One may justify the calculation first in the upper half-plane by Fubini's theorem and then take boundary values. For $a_t < x < 1$, the upper limit in (3.4) is t, and the integral is strictly smaller than $Q'(x)$. Symmetry gives the corresponding statements on the negative half-axis. These identities prove the equilibrium equalities and inequalities. □

Put $A = [-1/4, 1/4]$. We will use the following consequences of [8].

Lemma 5 (Uniform weighted kernel estimates)**.** *Fix $0 < t < 1$ sufficiently close to one that $A \Subset (-a_t, a_t)$, and choose $a_t < b < 1$. There are intervals*

$$A \Subset J_0 \Subset J_1 \Subset (-b, b)$$

and a bounded complex neighborhood Ω of $[-b,b]$, with $\overline{\Omega} \subset \{z : |\operatorname{Re} z| < 1\}$, and $\delta > 0$ with the following property. Let $\mathcal{U}$ consist of the bounded holomorphic functions F on Ω, real on $\Omega \cap \mathbb{R}$, for which $\|F - Q\|_\Omega < \delta$, where $\|F\|_\Omega = \sup_{z\in\Omega}|F(z)|$. Let $s \to \infty$, $m = \lfloor ts \rfloor$, and let Q_s belong to $\mathcal{U}$. If $p_0, \dots, p_{m-1}$ are orthonormal for $e^{-2sQ_s(x)}\,dx$ on $[-b,b]$, set

$$K_s(x,z) = e^{-sQ_s(x) - sQ_s(z)} \sum_{j=0}^{m-1} p_j(x)p_j(z), \qquad c_s = \frac{s}{2}. \tag{3.5}$$

Let $\beta_{s,y}$ be twice the density at y of the equilibrium measure of mass m/s for Q_s. Then, uniformly for the indicated family of external fields,

$$\frac{K_s(y+u/c_s, y+v/c_s)}{c_s} = \frac{\sin(\pi\beta_{s,y}(u-v))}{\pi(u-v)} + o(1), \tag{3.6}$$

$$\frac{|K_s(x,z)|}{c_s} \le \frac{C}{1+s|x-z|} \qquad (x,z \in J_1), \tag{3.7}$$

$$\frac{|K_s(x,z)|}{c_s} \le Cs^{-5/6} \qquad (x \in [-b,b]\setminus J_0,\ z \in A + O(s^{-1})). \tag{3.8}$$

In (3.6)*, $y \in A$, u,v range over bounded sets, and the quotient at $u=v$ is interpreted by continuity.*

Proof. In the notation of [8], take

$$N = m, \qquad J = [-b,b], \qquad V = V_s := \frac{2s}{m}Q_s.$$

Then $e^{-mV_s} = e^{-2sQ_s}$, and their kernel $K_{N,V}$ is exactly K_s. The base field $V_0 = 2Q/t$ satisfies their assumptions $(\mathrm{GA})_2$: it is analytic near the compact interval J, has $V_0'' \ge 2/t > 0$, and its equilibrium probability measure is ρ_t/t, supported on $[-a_t, a_t] \Subset J$.

Choose Ω as in [8, Remark 1.1] and a fixed sup-norm neighborhood of V_0 supplied by their Theorem 1.3(b). Since $m/s \to t$, decreasing δ and then taking s large ensures that every V_s lies in this same neighborhood. By their Lemma 2.4 the support endpoints depend continuously on V. Thus J_0, J_1 can be chosen inside every support, with fixed positive distances between A, their boundaries, the support endpoints, and $\pm b$. Cauchy's estimates also preserve a positive lower bound for V_s''. In particular, the density factor G_V in their (1.14)–(1.15) stays positive and bounded on the corresponding rescaled intervals.

Let ψ_s be the equilibrium probability density for V_s. Their affine map λ_{V_s} takes $[-1,1]$ onto the equilibrium support, and their rescaled density satisfies

$$\rho_{V_s}(\xi) = \lambda_{V_s}'\, \psi_s(\lambda_{V_s}(\xi)).$$

Then $\rho_s = (m/s)\psi_s$ is the equilibrium density of mass m/s for Q_s, and

$$m\psi_s(y) = s\rho_s(y) = c_s\beta_{s,y}.$$

In [8, Theorem 1.8], take the base point $\lambda_{V_s}^{-1}(y)$ and microscopic variables $\beta_{s,y}u$, $\beta_{s,y}v$. The preceding identities give (3.6), with the uniform error in part (b) of that theorem.

For (3.7), the bulk vector k in [8, Theorem 1.3(a), (1.25)] is uniformly bounded. The difference quotient therefore gives $O(|x-z|^{-1})$ when $|x-z| \ge s^{-1}$. For $|x-z| < s^{-1}$, the diagonal bound $K_s(x,x) = O(s)$ from Theorem 1.8 and Cauchy–Schwarz give the required $O(s)$ bound. For (3.8), the endpoint components of the vector k in that theorem have the forms, up to bounded multipliers,

$$m^{1/6}\operatorname{Ai}(\zeta), \qquad m^{-1/6}\operatorname{Ai}'(\zeta), \qquad |\zeta| = O(m^{2/3}).$$

The real Airy bounds $|\operatorname{Ai}(\zeta)| \le C$ and $|\operatorname{Ai}'(\zeta)| \le C(1+|\zeta|)^{1/4}$ give $\|k\| = O(m^{1/6})$ there. The bulk and exterior formulas give bounded vectors away from the endpoints. At $z \in A + O(s^{-1})$ the vector is bounded, while $|x-z|$ is bounded below. Thus (1.25), including its $O(m^{-1})$ remainder, gives $|K_s(x,z)| = O(m^{1/6})$. Since $m \asymp s$ and $c_s = s/2$, this proves (3.8). The neighborhood and error bounds are uniform by Theorem 1.3(b); Lemma 2.4 and the endpoint formulas keep the affine rescaling and the multipliers uniformly bounded after $\mathcal{U}$ is fixed. $\square$

4 Local amplification with the external-node weight

Consider a sequence of finite node sets $Y \subset [-1,1]$, with $N = \#Y$, and fix

$$I = (x_0 - h, x_0 + h) \Subset (-1,1), \qquad s = \#(Y \cap I).$$

In the coordinate $u = (x - x_0)/h$, define

$$W(u) = \prod_{y \in Y \setminus I} (x_0 + hu - y), \qquad Q_s(u) = -\frac{1}{s} \log \left| \frac{W(u)}{W(0)} \right|. \tag{4.1}$$

The function Q_s is convex on $(-1,1)$.

Proposition 6. *For each $H > 1$, there are $\epsilon_H, c_H, \delta_H > 0$, $b_H \in (0,1)$, and a bounded complex neighborhood Ω_H of $[-b_H, b_H]$, with $\overline{\Omega}_H \subset \{z : |\operatorname{Re} z| < 1\}$, with the following property. Let $\mathcal{U}_H$ be the ball $\|F - Q\|_{\Omega_H} < \delta_H$ of bounded holomorphic functions on Ω_H that are real on $\Omega_H \cap \mathbb{R}$. For any fixed interval I and sequence of node sets Y as above, suppose that $s \to \infty$, that the normalized analytic continuation of Q_s belongs to $\mathcal{U}_H$, and that*

$$\#\{y \in Y : (y - x_0)/h \in A\} \le \frac{s}{4} + \epsilon_H s. \tag{4.2}$$

If $d \ge 0$ are integers with $d = o(s)$, then, for all sufficiently large s, there are data $v_y \in [-1,1]$, $y \in Y$, such that every $p \in \Pi_{N+d}$ interpolating these data satisfies

$$|\{x \in I : |p(x)| > H\}| \ge c_H |I|. \tag{4.3}$$

Proof. We fix the auxiliary parameters in the following order, all before letting $s \to \infty$. First choose ρ so large that

$$c' \log \rho > 10H,$$

where c' is the constant in Lemma 2, and set $\epsilon_H = 1/(16\rho)$, $R_0 = \rho - \sqrt{\rho}$, and $\gamma = R_0^{-1/9}$. Choose $t < 1$ so close to one that

$$\inf_{y \in A} 2\pi \rho_t(y) > \pi - 2\gamma.$$

Fix $K = 3$, and then choose $\kappa > 0$ so small that $t + \kappa < 1$ and the Jackson factor (4.8) below is larger than $9/10$ whenever $y \in A$ and $|c_s(u - y)| \le \rho + 1$, for all sufficiently large s. Choose b with $a_{t+\kappa} < b < 1$, choose $A \Subset J_0 \Subset J_1 \Subset (-b, b)$ inside the common bulk region, and shrink the analytic neighborhood of Q so that the kernel estimates of Section 3 are uniform there, the supports of the equilibrium measures of masses near $[t, t+\kappa]$ stay a fixed positive distance from $\pm b$, and

$$\inf_{y \in A} \pi \beta_{s,y} > \pi - 3\gamma. \tag{4.4}$$

These choices give $b_H = b$, Ω_H, δ_H, and $\mathcal{U}_H$ in the statement. Set $\eta = 1/10$. By the uniform integral-tail bound in Lemma 2, choose T so that the truncation error in (4.6) is less than $\eta/2$. Next choose $R > \max\{\rho + 1, 2T\}$. Finally choose $D > 2R$ so that

$$2C_\rho \sum_{j=0}^{\infty} \frac{1}{1 + R + jD} \, \frac{1}{(1 + \kappa(R + jD))^{2K}} < \frac{1}{10}. \tag{4.5}$$

Here C_ρ is the constant in (4.10); first increasing R and then D makes (4.5) hold. All these parameters depend only on H.

Partition the interval c_sA into boxes of length 2ρ, discarding incomplete boxes at the ends. There are $s/(8\rho) - O(1)$ boxes. Call a box admissible if it contains at most $2\rho + 1$ scaled nodes. If B is the number of admissible boxes, then

$$\left(\frac{s}{8\rho} - O(1) - B\right)(2\rho + 2) \le \frac{s}{4} + \epsilon_H s.$$

The choice of ϵ_H implies $B \ge c_\rho s$.

For an admissible box centered at $c_s y$, apply Lemma 2 to its translated nodes, obtaining g. For a fixed truncation parameter T, define

$$G_{s,y}(u) = \int_{-T}^{T} g(v) \frac{K_s(u, y + v/c_s)}{c_s}\, dv. \tag{4.6}$$

This is $e^{-sQ_s(u)}$ times a polynomial of degree at most $m-1$. By (4.4) and the Paley–Wiener theorem,

$$\int_{\mathbb{R}} g(v) \frac{\sin(\pi\beta_{s,y}(\xi - v))}{\pi(\xi - v)}\, dv = g(\xi).$$

With η, T, R fixed above, Equation (3.6) implies, for all sufficiently large s,

$$\sup_{|\xi|\le R} |G_{s,y}(y + \xi/c_s) - g(\xi)| < \eta. \tag{4.7}$$

The uniform L^1 bound makes this assertion uniform even when g varies with the node configuration.

To obtain summable tails, multiply by an algebraic Jackson factor. Write $u = \cos\theta$, $y = \cos\alpha$, and, with $K = 3$ and κ fixed above, set $k = \lfloor \kappa s/(2K) \rfloor$. Define

$$\begin{aligned} D_k(v) &= \frac{\sin((k + 1/2)v)}{(2k+1)\sin(v/2)}, \\ L_{s,y}(u) &= D_k(\theta - \alpha)^{2K} + D_k(\theta + \alpha)^{2K}. \end{aligned} \tag{4.8}$$

This expression is even in θ, so it is an algebraic polynomial in u of degree at most $2Kk$. Since $y \in A$, it is bounded on $[-1, 1]$ by $1 + o(1)$. By the choice of κ, it is greater than $9/10$ throughout the scaled neighborhood containing the peak interval of g. Moreover,

$$|L_{s,y}(u)| \le C(1 + \kappa s|u - y|)^{-2K} + Cs^{-2K}. \tag{4.9}$$

Put $F_{s,y} = G_{s,y}L_{s,y}$. For $u \in J_1$ with $|c_s(u - y)| > 2T$, (3.7) gives

$$\begin{aligned} |F_{s,y}(u)| \le\ & C_\rho(1 + |c_s(u - y)|)^{-1}(1 + \kappa|c_s(u - y)|)^{-2K} \\ & + O_\rho(s^{-2K}). \end{aligned} \tag{4.10}$$

For $u \in [-b, b] \setminus J_0$, (3.8) and (4.9) give

$$|F_{s,y}(u)| = O_\rho(s^{-2K-5/6}). \tag{4.11}$$

To treat $b < |u| < 1$, write

$$F_{s,y}(u) = e^{-sQ_s(u)}H_{s,y}(u), \qquad \deg H_{s,y} \le D_s := m - 1 + 2Kk < s.$$

Let σ be the equilibrium measure of mass D_s/s for the field Q_s on $[-b, b]$. Its support remains in a fixed compact subinterval of $(-b, b)$. This follows from the continuity of the support endpoints in [8, Lemma 2.4], after shrinking the external-field neighborhood to cover masses near both t and $t+\kappa$. Let F be the constant value of $Q_s - V_\sigma$ on $\operatorname{supp}\sigma$. Since $\deg H_{s,y} \le D_s = s\sigma(\mathbb{R})$, the maximum principle applied to $\log|H_{s,y}| - sV_\sigma$ on the complement of the support, including infinity, yields

$$|e^{-sQ_s(u)} H_{s,y}(u)| \le \left\| e^{-sQ_s} H_{s,y} \right\|_{\operatorname{supp}\sigma} e^{-s(Q_s(u)-V_\sigma(u)-F)}. \tag{4.12}$$

For $u \in \operatorname{supp}\sigma \cap J_0$, the points $y + v/c_s$, $|v| \le T$, lie in J_1 for all large s. Hence (3.7), the uniform L^1-bound on g, and $\|L_{s,y}\|_\infty \le 1 + o(1)$ give

$$|F_{s,y}(u)| \le C_\rho.$$

For $u \in \operatorname{supp}\sigma \setminus J_0$, the same conclusion follows from (3.8), since $y + v/c_s$ stays within $O(s^{-1})$ of A. Consequently

$$\left\| e^{-sQ_s} H_{s,y} \right\|_{\operatorname{supp}\sigma} = \|F_{s,y}\|_{\operatorname{supp}\sigma} \le C_\rho \tag{4.13}$$

uniformly in the admissible box and in the allowed external field.

Put $D_s^*(u) = Q_s(u) - V_\sigma(u) - F$. The function Q_s is convex on $(-1,1)$, while $V_\sigma'' < 0$ off $\operatorname{supp}\sigma$. After shrinking the analytic neighborhood of Q, there is $c_0 > 0$ such that $Q_s'' \ge c_0$ on $[-b,b]$, and hence

$$(D_s^*)'' = Q_s'' - V_\sigma'' \ge c_0 \qquad \text{off } \operatorname{supp}\sigma \text{ in } [-b,b].$$

Let α_s and β_s be the leftmost and rightmost points of $\operatorname{supp}\sigma$. The support remains a fixed positive distance from $\pm b$. Since $D_s^* \ge 0$ on $[-b,b]$ and $D_s^* = 0$ on the support, its outward one-sided derivative at α_s and β_s is nonnegative. Integrating the preceding inequality from the support to $\pm b$ gives, for some $c > 0$ independent of s,

$$D_s^*(-b) \ge c, \qquad D_s^*(b) \ge c.$$

Outside $[-b,b]$, strict convexity on the two complementary intervals shows that D_s^* continues to increase toward the endpoints. Therefore (4.12) and (4.13) give

$$|F_{s,y}(u)| = O_\rho(e^{-cs}) \qquad (b < |u| < 1). \tag{4.14}$$

Retain admissible boxes whose centers in the $c_s u$ coordinate are separated by at least the fixed distance $D > 2R$. By (4.5), the sum of the tails in (4.10) from all centers outside the R-window is less than $1/10$. One residue class of the box indices retains at least $c_{\rho,D}s$ admissible boxes. At any point there is at most one center within scaled distance R. In this window use (4.7). At nodes of the same box use $|g| \le 1$; outside the box use (2.3). The remaining terms are controlled by the absolutely summable tails (4.10)–(4.14). The $O_\rho(s^{-2K})$ remainders remain $o(1)$ after summing over all retained boxes. Thus

$$\max_{z \in Y \cap I} \sum_j \left| F_{s,y_j}\left(\frac{z - x_0}{h}\right) \right| \le 2. \tag{4.15}$$

On $(-1,1)$, the ratio $W(u)/W(0)$ is positive, and hence $e^{-sQ_s(u)} = W(u)/W(0)$. Each function

$$B_j(x) = F_{s,y_j}\left(\frac{x - x_0}{h}\right)$$

therefore extends to a polynomial in the original coordinate x, vanishes at $Y \setminus I$, and has degree at most

$$N - s + m - 1 + 2Kk \le N - 1. \tag{4.16}$$

On its peak interval, (4.7) and the choice of κ give $F_{s,y_j} \geq \frac{9}{10}(10H - \eta)$. The other retained centers contribute at most $1/10 + o(1)$ there. Thus the B_j satisfy the peak dominance and node bounds in Lemma 3, on node-free intervals of length at least ch/s in the original coordinate. There are at least $c_{\rho,D}s$ such intervals. Since $d = o(s)$, that lemma supplies data for which every interpolant satisfies $|p| > H$ on a fixed fraction of them, proving (4.3). $\square$

5 Intervals with small node density

Proposition 7 (Low-density case)**.** *For each $H > 1$, there is $c_H > 0$ with the following property. Fix $I \Subset (-1, 1)$, and let $I^\theta = \arccos I \subset (0, \pi)$. Suppose that*

$$\frac{\max_{\theta \in \overline{I^\theta}} \sin\theta}{\min_{\theta \in \overline{I^\theta}} \sin\theta} \leq 2.$$

Let $Y \subset [-1, 1]$ be a sequence of sets with $N = \#Y \to \infty$ satisfying

$$\#(Y \cap I) \leq \frac{N|I^\theta|}{2\pi}, \tag{5.1}$$

and let $d \geq 0$ be integers with $d = o(N)$. For all sufficiently large N, there are data $v_y \in [-1, 1]$ on Y such that every $p \in \Pi_{N+d}$ interpolating them satisfies

$$|\{x \in I : |p(x)| > H\}| \geq c_H |I|. \tag{5.2}$$

Proof. Write $x = \cos\theta$ and $v = N\theta/\pi$. Choose ρ so that $c' \log\rho > 10H$, and then choose $D > 4\rho$ so that

$$2C \sum_{j=1}^{\infty} (jD - 2\rho)^{-3} < \frac{1}{10}. \tag{5.3}$$

Partition the corresponding v-interval into boxes of length 2ρ, discarding the incomplete boxes at the two ends. Let M be the number of complete boxes and B_{bad} the number containing at least $2\rho + 2$ nodes. The v-interval has length $L = N|I^\theta|/\pi$, so (5.1) gives

$$B_{\text{bad}}(2\rho + 2) \leq \frac{L}{2}, \qquad L < 2\rho(M + 2).$$

Thus

$$B_{\text{bad}} < \frac{\rho(M + 2)}{2\rho + 2} \leq \frac{M}{2} + 1.$$

Hence at least $M/2 - 1$ complete boxes contain at most $2\rho + 1$ nodes; in particular, this is at least $M/3$ once $M \geq 6$. Retain such boxes with a fixed separation D, and apply Lemma 2 at their centers a_j. For the resulting functions g_j, define

$$\Phi_{N,j}(\theta) = \sum_{\ell \in \mathbb{Z}} \left[g_j\left(\frac{N\theta}{\pi} - a_j + 2N\ell\right) + g_j\left(-\frac{N\theta}{\pi} - a_j + 2N\ell\right) \right]. \tag{5.4}$$

The series converges uniformly by (2.3). Its Fourier coefficients vanish at all frequencies k for which

$$\frac{\pi|k|}{N} > \pi - 4\gamma,$$

by the Paley–Wiener theorem. It is therefore an even real trigonometric polynomial of degree less than N, and hence an algebraic polynomial of degree less than N in $x = \cos\theta$.

Because I^θ stays away from 0 and π, the reflected centers stay a fixed angular distance from $[0, \pi]$, modulo 2π. Their contribution, and that of every nonzero period of the unreflected term, is $O(N^{-3})$ uniformly on $[0, \pi]$, by (2.3). Summing these errors over $O(N)$ retained boxes gives $o(1)$.

By (2.3) and (5.3), the sum of the principal tails from all other retained boxes is less than $1/10$. For each retained box let

$$B_{N,j}(x) = \Phi_{N,j}(\arccos x).$$

If a node $z = \cos\theta$ lies in the j-th retained box, the principal term of $B_{N,j}(z)$ has absolute value at most one by (2.1); if it lies outside that box, the same term is controlled by (2.3). The reflected and nonzero-period terms contribute only $o(1)$ after summation. Therefore, for all sufficiently large N,

$$\max_{z\in Y} \sum_j |B_{N,j}(z)| \le 2. \tag{5.5}$$

Each retained box contains a node-free interval of a fixed positive length in the v-coordinate where its principal function is at least $c' \log\rho > 10H$. The sum of all other terms is at most $1/10 + o(1)$, so these intervals satisfy the peak dominance condition of Lemma 3. There are at least $c_{\rho,D} N |I^\theta|$ such intervals, each of angular length at least c/N. Since $d = o(N)$, that lemma supplies bounded data forcing every interpolant to satisfy $|p| > H$ on a fixed fraction of them. The assumed ratio bound for $\sin\theta$ converts their total angular length into at least $c_H|I|$ in the original coordinate, proving (5.2). □

6 Local intervals for an arbitrary node array

Fix $H > 1$. Throughout this section, $S \subset [-1, 1]$ denotes an arbitrary fixed finite set, and $Y_n = X_n \setminus S$. Pass to a subsequence, with indices still denoted by n, such that

$$\nu_n = \frac{1}{n} \sum_{x\in X_n} \delta_x \implies \nu. \tag{6.1}$$

Only this subsequence will be used in the construction. Removing the nodes belonging to any fixed finite set does not alter the weak limit. Put

$$V = V_\nu, \qquad m_0 = \inf_{\operatorname{supp}\nu} V.$$

The function V belongs to $L^1[-1, 1]$, and $V \ge m_0$ throughout $[-1, 1]$. To see the latter assertion off the support, note that V is concave on each bounded complementary interval and has the corresponding endpoint limits. On the two exterior intervals it is monotone toward the support.

6.1 Points above the minimum potential

Put $P_n = P_{Y_n}$. For $z \in \operatorname{supp}\nu$, choose $z_n \in Y_n$ with $z_n \to z$. Truncation of the logarithmic kernel gives

$$\limsup_{n\to\infty} \frac{1}{n} \log |P_n'(z_n)| \le V(z), \qquad \frac{1}{n} \log |P_n| \longrightarrow V \quad \text{in } L^1[-1, 1]. \tag{6.2}$$

For the first assertion, truncate $\log|x - y|$ from below at $-A$. The contribution of the omitted diagonal term is A/n; one first takes the weak limit and then lets $A \to \infty$. For the second, use the continuity of

$$y \longmapsto (x \longmapsto \log|x - y|)$$

as a map from $[-1,1]$ into $L^1[-1,1]$.

Suppose first that $V(z) > -\infty$, and let

$$A_{z,\eta} = \{x : V(x) > V(z) + 3\eta\}, \qquad \eta > 0.$$

Outside a subset of $A_{z,\eta}$ whose measure tends to zero, (6.2) implies

$$|\ell_n(x)| \geq e^{\eta n}, \qquad \ell_n(x) = \frac{P_n(x)}{(x - z_n)P_n'(z_n)}. \tag{6.3}$$

Specify the value one at z_n, and zero at all other nodes of Y_n. Every polynomial of degree at most $n + r_n$ satisfying these data can be written as

$$p = \ell_n T, \qquad T(z_n) = 1, \qquad \deg T \leq r_n + \#S + 1 = o(n). \tag{6.4}$$

If $|p| \leq H$ on a subset of $A_{z,\eta}$ of fixed positive measure, (6.3) and Remez's inequality [7, p. 93, (3)–(4)] yield

$$|T(z_n)| \leq H e^{-\eta n} C^{r_n + \#S + 1} < 1$$

for all sufficiently large n, a contradiction. Here C depends only on the fixed measure lower bound. Thus the measure of the low set in $A_{z,\eta}$ tends to zero uniformly over all the polynomials in (6.4). At every density point of $A_{z,\eta}$, each sufficiently small fixed interval therefore admits data forcing large values on at least half its length.

Choose countably many z for which $V(z)$ decreases to m_0, and take positive rational η. The sets $A_{z,\eta}$ cover $\{V > m_0\}$ when the chosen source potentials are finite. If some $V(z) = -\infty$, the first assertion of (6.2) instead gives

$$\frac{1}{n} \log |P_n'(z_n)| \longrightarrow -\infty.$$

On each set $\{V \geq -A\}$, the same argument gives (6.3) for any fixed positive exponential rate, outside a set of vanishing measure. Letting $A \to \infty$ covers almost all of $[-1,1]$. In particular, if $m_0 = -\infty$, this subsection supplies local intervals at almost every point.

6.2 The minimum-potential set

Suppose that $m_0 > -\infty$, and let

$$E = \{x : V(x) = m_0\}.$$

By the differentiation theorem for measures, ν has a finite Lebesgue density at almost every point. At points where this density is zero, all sufficiently small intervals satisfy the strict limiting form of (5.1). Fixing such an interval before taking $n \to \infty$, and choosing endpoints that are not atoms of ν, allows Proposition 7 to be applied.

We isolate the local field arising at the remaining points.

Lemma 8 (Local outer field)**.** *Let x_0 be a density point of E and a Lebesgue point of the absolutely continuous part of ν, with finite positive density w, and suppose that the singular part of ν has density zero at x_0. Put*

$$I_h = (x_0 - h, x_0 + h), \qquad a_h = \nu(I_h),$$

and let μ_h be the image of $\nu|_{I_h}/a_h$ under $x \mapsto (x - x_0)/h$. Then, as $h \downarrow 0$,

$$a_h \sim 2wh, \qquad \mu_h \Longrightarrow \frac{du}{2},$$

and

$$V_{\mu_h} \longrightarrow Q-1 \quad \text{in } L^1(-1,1). \tag{6.5}$$

Define the normalized outer potential

$$A_h(u) = -\frac{1}{a_h}\int_{\mathbb{R}\setminus I_h} \log|x_0+hu-y|\,d\nu(y).$$

Then

$$A_h(u)-A_h(0) \longrightarrow Q(u) \quad \text{locally uniformly on } (-1,1). \tag{6.6}$$

Moreover, the functions $A_h - A_h(0)$ admit analytic continuations $\widetilde{A}_h$ to the strip $\Omega_0 = \{z\in\mathbb{C} : |\operatorname{Re} z|<1\}$, and

$$\widetilde{A}_h \longrightarrow Q$$

locally uniformly there.

Proof. The differentiation theorem gives $a_h \sim 2wh$ and $\mu_h \Rightarrow du/2$. The continuity of

$$y \longmapsto (x \longmapsto \log|x-y|)$$

from $[-1,1]$ into $L^1[-1,1]$ gives (6.5), since the logarithmic potential of $du/2$ is $Q-1$.

For u corresponding to $E\cap I_h$, the equality $V=m_0$ gives

$$A_h(u)+\frac{m_0}{a_h}-\log h = V_{\mu_h}(u). \tag{6.7}$$

The relative Lebesgue measure of this set in $(-1,1)$ tends to one, and the left side of (6.7) is convex. Thus (6.5) and (6.7) imply convergence in measure to $Q-1$. Finite convex functions that converge in measure to a continuous finite convex function converge uniformly on compact subintervals: convergence points on either side of a compact interval bound the secant slopes, giving equicontinuity. This proves (6.6).

Writing $\xi=(y-x_0)/h$, one has $|\xi|\ge 1$ for $y\notin I_h$. Define $\widetilde{A}_h$ on Ω_0 by $\widetilde{A}_h(0)=0$ and

$$\widetilde{A}_h'(z) = -\frac{h}{a_h}\int_{\mathbb{R}\setminus I_h} \frac{d\nu(y)}{x_0+hz-y}.$$

For real $u\in(-1,1)$, integration along the real segment gives $\widetilde{A}_h(u)=A_h(u)-A_h(0)$, and

$$\widetilde{A}_h''(z) = \frac{1}{a_h}\int_{\mathbb{R}\setminus I_h} \frac{d\nu(y)}{(z-\xi)^2}.$$

Fix a compact subset K of Ω_0, and choose a compact interval $K_1 \Subset (-1,1)$ whose interior contains $\operatorname{Re} K\cup\{0\}$. For sufficiently small fixed $\ell>0$, the kernels $(u-\xi)^{-2}$ are uniformly comparable on $[u-\ell,u+\ell]$, uniformly for $u\in K_1$. Hence

$$A_h''(u) \le C_\ell \int_{u-\ell}^{u+\ell} A_h''(v)\,dv = C_\ell\big(A_h'(u+\ell)-A_h'(u-\ell)\big).$$

The local uniform convergence in (6.6) and convexity give uniform interior bounds for A_h', hence for A_h'' on K_1. Since

$$|z-\xi|^{-2} \le |\operatorname{Re} z-\xi|^{-2},$$

$\widetilde{A}_h''$ is uniformly bounded for $\operatorname{Re} z\in K_1$. Integrating from zero and using the interior first-derivative bound gives local boundedness on Ω_0, hence a normal family. Every subsequential limit agrees with Q on the real interval by (6.6), and therefore equals Q by the identity theorem. $\square$

Lemma 9 (Discrete outer-field approximation). *Fix $h > 0$ such that the endpoints of I_h and the preimages of the endpoints of A are not atoms of ν, and put $s_n = \#(Y_n \cap I_h)$. Then*

$$\frac{s_n}{n} \longrightarrow a_h, \qquad \frac{\#\{y \in Y_n : (y - x_0)/h \in A\}}{s_n} \longrightarrow \mu_h(A). \tag{6.8}$$

Let $\mathcal{Q}_n$ be the normalized analytic continuation of the discrete field (4.1), with $\mathcal{Q}_n(0) = 0$. On every fixed complex neighborhood compactly contained in the domain of $\widetilde{A}_h$,

$$\mathcal{Q}_n \longrightarrow \widetilde{A}_h$$

locally uniformly. If $N_n = \#Y_n$, then the degree bound $n + r_n$ is $N_n + d_n$, where

$$d_n = r_n + \#(X_n \cap S) = o(s_n). \tag{6.9}$$

Proof. The two limits in (6.8) follow from weak convergence and the choice of the interval endpoints. Moreover,

$$\mathcal{Q}_n'(z) = -\frac{h}{s_n} \sum_{y \in Y_n \setminus I_h} \frac{1}{x_0 + hz - y}.$$

On a fixed complex neighborhood as above, the summands are uniformly bounded and equicontinuous as functions of the outer node. Since $n/s_n \to 1/a_h$, weak convergence of the empirical measures gives $\mathcal{Q}_n' \to \widetilde{A}_h'$ locally uniformly. Integrating from zero gives the asserted convergence of the fields. Finally, $r_n = o(n)$ and $s_n/n \to a_h > 0$ give (6.9). □

Now let x_0 satisfy the hypotheses of Lemma 8; these conditions hold at almost every point of E where the density of ν is positive. Since $\overline{\Omega}_H \Subset \Omega_0$, Lemma 8 gives $\left\|\widetilde{A}_h - Q\right\|_{\Omega_H} \to 0$. Choose h so small that this norm is less than $\delta_H/2$, for the neighborhood $\mathcal{U}_H$ in Proposition 6, and

$$\mu_h(A) < \frac{1}{4} + \frac{\epsilon_H}{2}.$$

This is possible because $\mu_h \Rightarrow du/2$ and $(du/2)(A) = 1/4$. We may also choose h so that the interval endpoints required in Lemma 9 are not atoms of ν. Lemma 9 gives $\left\|\mathcal{Q}_n - \widetilde{A}_h\right\|_{\Omega_H} < \delta_H/2$ for all sufficiently large n, so $\mathcal{Q}_n \in \mathcal{U}_H$. It also gives (4.2) and an excess degree $d_n = o(s_n)$. Therefore Proposition 6 applies.

Proposition 10 (Local forcing for arbitrary arrays). *For each $H > 1$, there is $c_H > 0$ such that every array (X_n) and sequence (r_n) as in Theorem 1 admit a subsequence with the following property. For almost every $x \in (-1, 1)$, there are arbitrarily small intervals $I \Subset (-1, 1)$ containing x such that, for every fixed finite $S \subset [-1, 1]$, all sufficiently late rows of the subsequence admit data bounded by one on $X_n \setminus S$ forcing*

$$|\{y \in I : |p_n(y)| > H\}| \geq c_H |I|$$

for every interpolant of degree at most $n + r_n$.

Proof. Use the weakly convergent subsequence in (6.1). Section 6.1 gives the assertion on $\{V > m_0\}$, with fraction 1/2. On $\{V = m_0\}$, Section 6.2 applies Proposition 7 where the density of ν is zero, and Proposition 6 where it is positive and finite. These cases exhaust almost every point. Take the minimum of their three constants. The intervals are selected from ν and H; for every fixed S, the same limits apply to $X_n \setminus S$, with excess degree $r_n + \#(X_n \cap S)$. □

7 A finite construction for common low sets

Proposition 11. *Fix $H > 1$, $\delta > 0$, and $N_0 \geq 1$. There are finitely many indices $n_1, \ldots, n_k \geq N_0$ and a function $g \in C([-1,1];\mathbb{R})$, with $\|g\|_\infty \leq 1$, such that every choice of polynomials*

$$p_j \in \Pi_{n_j + r_{n_j}}, \qquad p_j = g \text{ on } X_{n_j},$$

satisfies

$$\left| \bigcap_{j=1}^{k} \{x : |p_j(x)| \leq H\} \right| < \delta. \tag{7.1}$$

Proof. For any interval $J \subset (-1,1)$, the intervals furnished by Proposition 10 form a Vitali cover of almost all of J. Choose finitely many disjoint such intervals whose total length is at least $|J|/2$. For each interval, choose a sufficiently late new row of the fixed subsequence, assigning values only at nodes whose values have not yet been specified, applying Proposition 10 with S equal to the finite set of previously assigned nodes. For all choices of the resulting interpolating polynomials, the union of their sets of values exceeding H occupies at least

$$c_*|J|, \qquad c_* = \frac{c_H}{2} > 0, \tag{7.2}$$

inside J.

Assume $0 < \delta \leq 2$, since larger δ is immediate. Suppose that finitely many rows have already been chosen, with degree bounds $d_1, \ldots, d_k$. For every admissible choice of their polynomials, the common low set

$$L = \bigcap_{j=1}^{k} \{x : |p_j(x)| \leq H\}$$

has at most $2 + 2\sum_j d_j$ boundary points in $[-1,1]$. A constant polynomial equal to H or $-H$ introduces no additional boundary points. Choose a fixed partition of $[-1,1]$ into sufficiently short intervals that the cells meeting these boundary points have total length less than $\delta/4$. The same partition works for every admissible choice of the polynomials, since the boundary bound depends only on their degrees.

In each cell, carry out the construction giving (7.2), using new rows and assigning only new node values. If $|L| \geq \delta$, the cells contained entirely in L have total length at least

$$|L| - \delta/4 \geq 3|L|/4.$$

For the new common low set this gives

$$|L_{\text{new}}| \leq (1 - 3c_*/4)|L| \qquad \text{whenever } |L| \geq \delta. \tag{7.3}$$

Starting from $L = [-1,1]$, perform a finite number q of batches, where

$$2(1 - 3c_*/4)^q < \delta.$$

At each batch the new partition is chosen using the degree bounds of the rows already selected. Equation (7.3) shows that the final common low set has measure less than δ, uniformly over all admissible polynomial choices. All rows may be chosen beyond N_0.

The assigned values define a function on a finite union of nodes, with absolute value at most one. Extend it by linear interpolation between consecutive nodes and by constants to the endpoints of $[-1,1]$. The resulting continuous function g has $\|g\|_\infty \leq 1$ and satisfies (7.1). $\square$

8 Proof of Theorem 1

Work in the real Banach space $C([-1,1];\mathbb{R})$. For $H > 0$ and positive integers N, k, let $\mathcal{O}_{H,N,k}$ consist of the functions f for which there is a finite set $S \subset \{N, N+1, \dots\}$ such that every choice

$$p_n \in \Pi_{n+r_n}, \qquad p_n = f \text{ on } X_n \quad (n \in S),$$

satisfies

$$\left| \bigcap_{n \in S} \{x : |p_n(x)| \le H\} \right| < \frac{1}{k}.$$

Lemma 12. *If $H > M > 0$, then $\mathcal{O}_{H,N,k} \subset \operatorname{int} \mathcal{O}_{M,N,k}$.*

Proof. Let $f \in \mathcal{O}_{H,N,k}$, with finite set S as above. For each $n \in S$, let L_n be the Lagrange interpolation operator on X_n, and write $\ell_{n,\xi}$ for its fundamental polynomials. Put

$$C = 1 + \sum_{n \in S} \sum_{\xi \in X_n} \|\ell_{n,\xi}\|_\infty .$$

If $\|g - f\|_\infty < (H - M)/C$ and p_n interpolates g, then $q_n = p_n - L_n(g - f)$ interpolates f with the same degree bound. Moreover,

$$\|L_n(g - f)\|_\infty < H - M, \qquad \{|p_n| \le M\} \subset \{|q_n| \le H\}.$$

The defining bound for f therefore gives the defining bound for g at height M. Thus the ball of radius $(H - M)/C$ about f lies in $\mathcal{O}_{M,N,k}$. □

Lemma 13. *For every positive integer triple M, N, k, the open set $\operatorname{int} \mathcal{O}_{M,N,k}$ is dense.*

Proof. Let an open ball in $C([-1,1];\mathbb{R})$ be given. By polynomial approximation, it contains a polynomial h. Choose $\varepsilon > 0$ so that $h + \varepsilon g$ belongs to the ball whenever $\|g\|_\infty \le 1$. Apply Proposition 11 with

$$H > \max\left\{1, \frac{M + 1 + \|h\|_\infty}{\varepsilon}\right\}, \qquad \delta = \frac{1}{k},$$

requiring all selected indices to be at least N and large enough that $n + r_n \ge \deg h$. Let g be the resulting function and set $f = h + \varepsilon g$. For any admissible interpolant p_n of f on a selected row,

$$q_n = \frac{p_n - h}{\varepsilon}$$

is an admissible interpolant of g, and $|p_n| \le M + 1$ implies $|q_n| < H$. Hence $f \in \mathcal{O}_{M+1,N,k}$. Lemma 12 puts f in $\operatorname{int} \mathcal{O}_{M,N,k}$, as required. □

Proof of Theorem 1. By Lemma 13 and the Baire category theorem, choose

$$f \in \bigcap_{M,N,k \ge 1} \operatorname{int} \mathcal{O}_{M,N,k}.$$

Fix any admissible interpolation sequence (p_n). For every M, N, k, the common tail low set

$$E_{M,N} = \bigcap_{n \ge N} \{x : |p_n(x)| \le M\}$$

is contained in the finite common low set supplied by $f \in \mathcal{O}_{M,N,k}$. Thus $|E_{M,N}| < 1/k$ for every k, and $|E_{M,N}| = 0$. Outside the null set $\bigcup_{M,N \ge 1} E_{M,N}$, every tail of $(|p_n(x)|)$ exceeds every positive integer threshold. This proves (1.1). □

Code availability

A partial Lean 4 formalization is available at `https://github.com/FireflySentinel/erdos-1152`. It covers the sign-change bound in Lemma 3, the scalar calculations of Lemma 4, the deduction from (6.3) using Remez's inequality, and the finite construction and Baire category argument in Sections 7–8. Remez's inequality and the remaining analytic estimates are hypotheses.

Use of generative AI

The author used GPT-5.6 and GPT-6 Astra in developing the arguments involving polynomial corrections, local external fields, weighted kernels and finite interpolation, and OpenAI Codex (GPT-6) for the Baire category argument and the Lean formalization. The author checked the arguments and their use of the cited results and is responsible for the content.

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